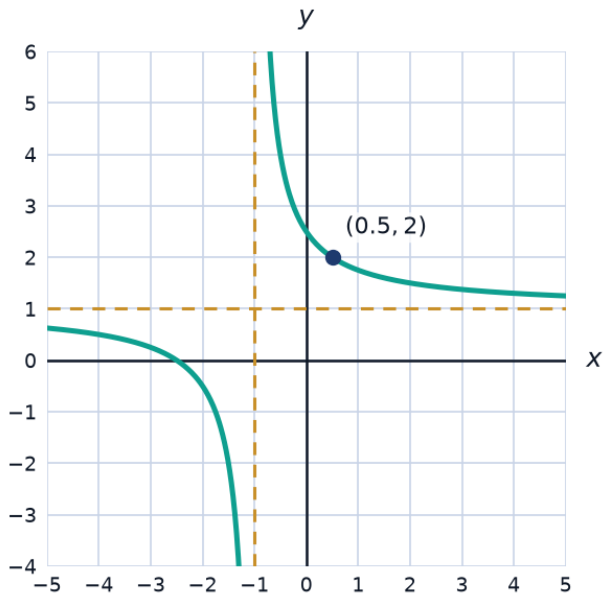
Limits

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

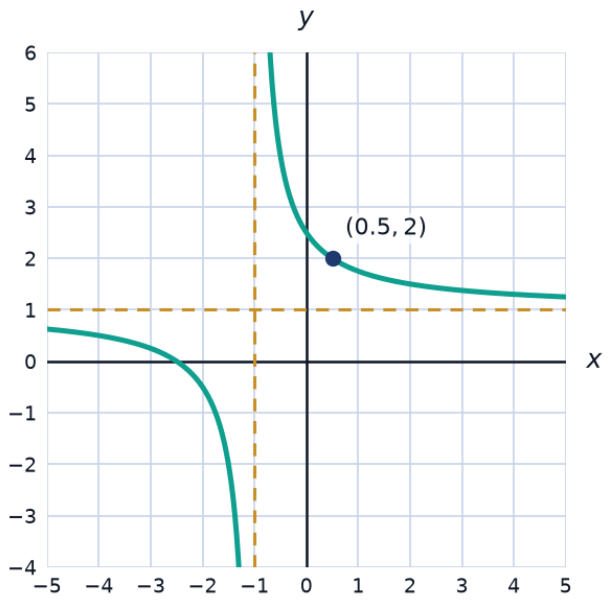
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find $\lim_{x \to -1} (x^3 + 4x)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

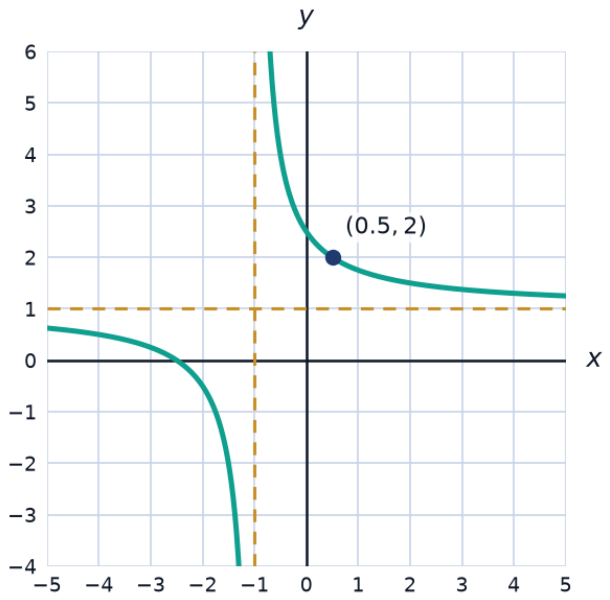
Find $\lim_{x \to \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.



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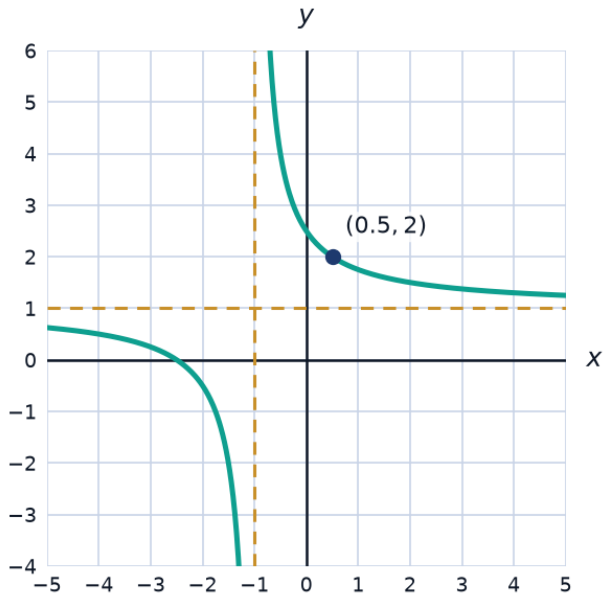
3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $z_1 = 5 - 2i$ and $z_2 = -3 + 7i$, find the value of $2z_1 - z_2$ in standard form $a + bi$.



4 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find $\lim_{x \to -9} \frac{x^2 + 14x + 45}{x + 9}$.



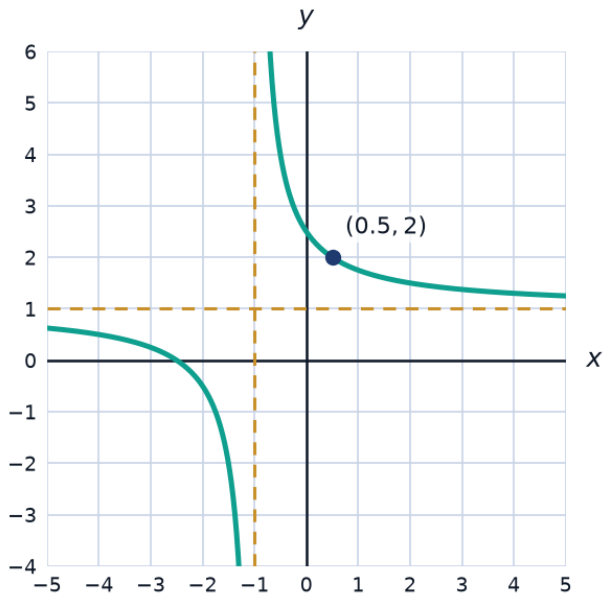
5 [Foundational] • Target: Direct substitution

Find $\lim_{x \to 3} (2x^2 - x + 1)$.

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6 [Foundational] • Target: Direct substitution

Find $\lim_{x \to -1} (x^3 + 4x)$.



7 [Foundational] • Target: Indeterminate forms: factor and cancel

Find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$.

9 [Intermediate] • Target: Indeterminate forms: factor and cancel

Explain why you cannot find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.

10 [Intermediate] • Target: Limits at infinity for rational functions

Find $\lim_{x \to \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.

11 [Intermediate] • Target: Limits at infinity for rational functions

Find $\lim_{x \to \infty} \frac{4x + 1}{2x^2 + 3}$.

12 [Intermediate] • Target: Limits at infinity for rational functions

Explain what happens to $\lim_{x \to \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find $\lim_{x \to 3} (2x^2 - x + 1)$.

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14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

Find $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$.

16 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 1.01 Complex Numbers**

Explain why you cannot find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.

17 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 2.01 Matrices & Vectors**

Find $\lim_{x \to \infty} \frac{4x + 1}{2x^2 + 3}$.

18 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 3.01 Sequences & Series**

Explain what happens to $\lim_{x \to \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 4.01 Advanced Functions & Equations**

In triangle ABC, sides are $a = 5$, $b = 6$, and $c = 7$. Find the exact value of $\cos A$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

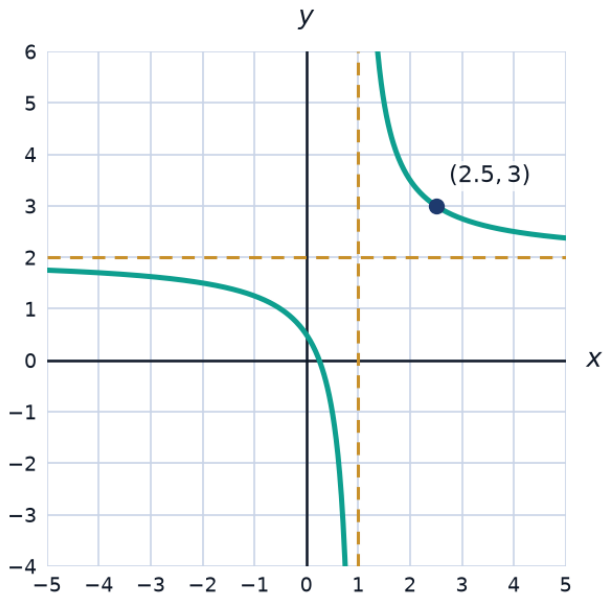
Let $f(x) = 3x - 1$ and $g(x) = 2x^2 + 5$. Find $f(g(3))$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find $\lim_{x \to 2} (-3x^2 + 7x - 7)$.



a. -3

b. -5

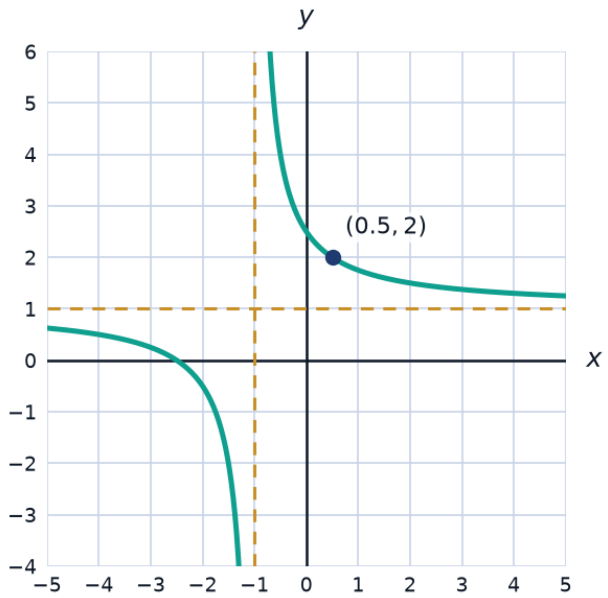
c. -7

d. 1

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find $\lim_{x \to 3} \frac{x^2 - 5x + 6}{x - 3}$.



a. 1

b. 3

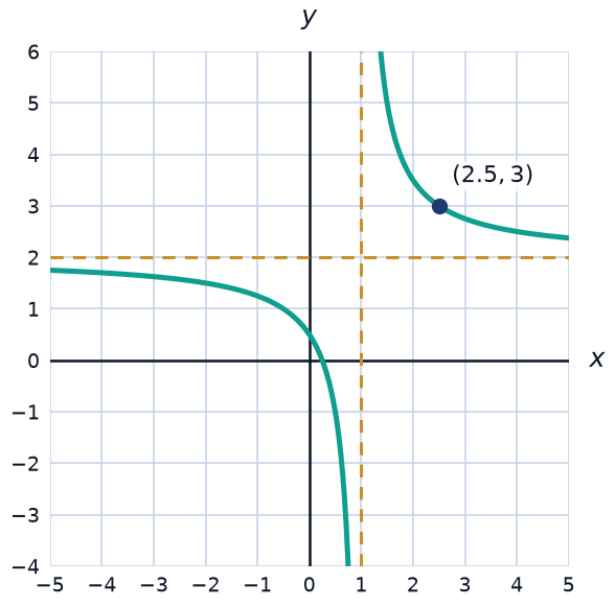
c. −1

d. 4

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find $\lim_{x \to \infty} \frac{-9x^3 - 8x}{-8x^2 + 2x}$ (state whether it is ∞ or $-\infty$).



a. ∞

b. $\frac{9}{8}$

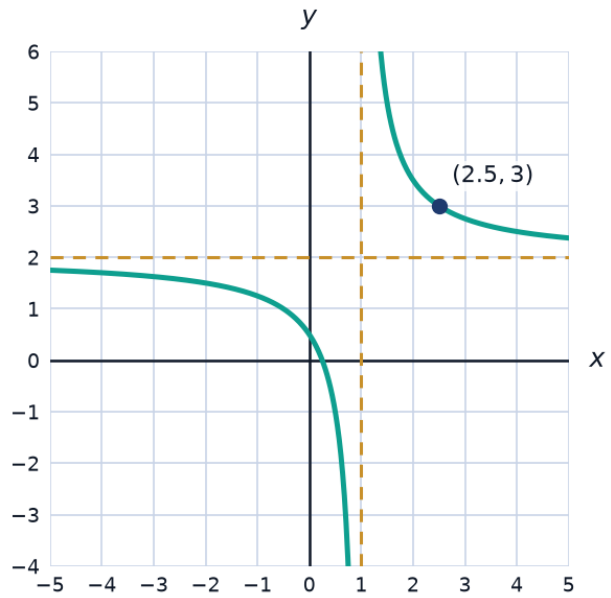
c. $-\infty$

d. 0

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is $\lim_{x \to 0^-} \frac{1}{x}$?



a. $-\infty$

b. ∞

c. 0

d. Does not exist

25 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find $\lim_{x \to -7} (3x^2)$.

a. 145

b. 147

c. -21

d. 149

26 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find $\lim_{x \to -6} (2x^2 + 5x - 9)$.

a. 35

b. 31

c. -51

d. 33

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find $\lim_{x \to -7} (4x^2 - 2x - 9)$.

a. 203

b. 201

c. 199

d. -23

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find $\lim_{x \to -4} (3x^2 + x + 4)$.

a. -12

b. 50

c. 46

d. 48

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find $\lim_{x \to 6} (2x^2 + 9x - 6)$.

a. 122

b. 120

c. 60

d. 118

30 [Intermediate] • Target: Direct substitution

Find $\lim_{x \to -3} (3x^2 - 8x - 2)$.

a. 13

b. 51

c. 49

d. 47

31 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find $\lim_{x \to -5} \frac{x^2 - 4x - 45}{x + 5}$.

a. 14

b. -16

c. -12

d. -14

32 [Intermediate] • Target: Limits at infinity for rational functions

Find $\lim_{x \to -2} \frac{x^2 + 3x + 2}{x + 2}$.

a. 2

b. -1

c. -3

d. 1

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33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find $\lim_{x \to -1} \frac{x^2 - 4x - 5}{x + 1}$.

a. -6

b. -8

c. 6

d. -4

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find $\lim_{x \to 7} \frac{x^2 - 16x + 63}{x - 7}$.

a. -2

b. -4

c. 0

d. 2

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find $\lim_{x \to -9} \frac{x^2 + 14x + 45}{x + 9}$.

a. -2

b. 4

c. -4

d. -6

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find $\lim_{x \to \infty} \frac{-7x^2 + 3x}{-5x^2 + 4x}$.

a. $\frac{7}{5}$

b. $-\frac{7}{5}$

c. $\frac{2}{5}$

d. $\frac{12}{5}$

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find $\lim_{x \to 0} \frac{\sqrt{x + 81} - \sqrt{81}}{x}$.

a. $\frac{1}{18}$

b. $\frac{1}{9}$

c. $\frac{1}{6}$

d. $\frac{2}{9}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

Find $\lim_{x\to 0} \frac{\sqrt{x+25}-\sqrt{25}}{x}$.

a. $\frac{2}{5}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$

d. $\frac{1}{5}$

39 [Challenging] • Target: Direct substitution • Cross-Lesson: 4.01 Advanced Functions & Equations

Find $\lim_{x\to\infty} \frac{-9x^3-8x}{-8x^2+2x}$ (state whether it is ∞ or $-\infty$).

a. ∞

b. $\frac{9}{8}$

c. $-\infty$

d. 0

40 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Find $\lim_{x\to\infty} \frac{2x^3-8x}{9x^2-3x}$ (state whether it is ∞ or $-\infty$).

a. 0

b. $\frac{2}{9}$

c. $-\infty$

d. ∞

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Complex Numbers

Find $\lim_{x\to 5} (-3x^2-9x+7)$.

a. -53

b. -113

c. -115

d. -111

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Matrices & Vectors

Find $\lim_{x\to -4} (-2x^2-4x+4)$.

a. -12

b. -10

c. -14

d. 28

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Sequences & Series

Find $\lim_{x \to 3} \frac{x^2 - x - 6}{x - 3}$.

a. 7

b. 5

c. 3

d. -5

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Functions & Equations

Find $\lim_{x \to 0} \frac{\sqrt{x + 25} - \sqrt{25}}{x}$.

a. $\frac{2}{5}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$

d. $\frac{1}{5}$

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Evaluate: $\lim_{x \to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$.

a. $-\frac{1}{4}$

b. $\frac{1}{4}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

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